**KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY**
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**Department Of Information Technology**

## MAJOR PROJECT STAGE – I

# DeciMind AI : An AI-Powered Decision Intelligence System

## ABSTRACT

TheModern decision-making systems in domains such as finance, healthcare, human resources, cybersecurity, and education increasingly rely on automated intelligence. However, many existing AI systems lack transparency, adaptability, and alignment with human judgment. Traditional rule- based systems are often rigid and unable to generalize across changing scenarios, while standalone machine learning models frequently function as black boxes with limited interpretability and governance. To address these limitations, this project introduces the Human Decision Intelligence System (HDIS), a dual-layer AI governance framework that combines rule-based reasoning with machine learning to support transparent, adaptive, and human-aligned decision-making. The system generates baseline decisions using domain-specific rule engines and enhances them with Logistic Regression models trained on historical decision logs to improve predictive alignment with human behavior.

HDIS incorporates a human-in-the-loop architecture that allows decisions to be reviewed, overridden, and recorded, enabling continuous learning and system evolution. Explainable AI (XAI) modules provide structured reasoning by identifying the key factors influencing each decision, improving trust and interpretability. An advanced analytics dashboard monitors performance through real-time metrics such as AI Trust Score, Override Rate, AI–ML Agreement, and System Reliability. The backend is built using Flask with JWT-based authentication, while the frontend is developed using React.js to create an interactive user experience. SQLite serves as the database for storing decision logs, which also act as the training dataset for machine learning models. By integrating governance mechanisms, explainability, and feedback-driven learning, HDIS establishes a scalable and trustworthy framework for deploying AI solutions across multiple real-world domains.

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